

Feature Build Spec

Authentication, Signup & RBAC

This document is written for the development team building HubblerX on Vibe/Cursor. It specifies exactly what fields appear on every signup and login screen, what each of the three roles (Student, College Admin, Support) can access, and the complete feature-to-role permission matrix. Read this before writing any auth, routing, or access-control code.

SYMBOL	MEANING
✓ Full	Full access to create, read, update, delete
■ Own only	Access to own data only (e.g. own college, own profile)
■ Read	Read/view only — no create or edit
■ Partial	Access to specific sub-features only — detailed in section
X None	No access — route is blocked, UI element hidden

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Student Signup & Login

The student signup is the highest-volume entry point. It must be fast on mobile web, require no app download, and complete in under 90 seconds. The flow uses a magic link (passwordless email) for login after the initial password is set at signup.

1.1 Student Signup Form — All Fields

FIELD	TYPE	REQ	VALIDATION	NOTES
Full Name	Text	Required	2–60 chars, letters + spaces only	Displayed on profile, certificate, leaderboard
Mobile Number	Phone	Required	10 digits, India (+91), OTP verified at signup	Primary login fallback, ticket delivery via SMS
College Name	Select	Required	Must match a college registered in HubblerX colleges table	Dropdown populated from verified colleges DB
College Location	Select	Required	Shown only if selected college has multiple campuses	e.g. SRM KTR vs SRM Vadapalani — child of college dropdown
Department	Select	Required	Dropdown from college's registered departments	Drives dept leaderboard and internship filters
Education Tier	Select	Required	One of: Diploma / UG / PG	Determines which leaderboard tier student is placed in
College Year Range	Year Range	Required	Start year + End year. e.g. 2022 – 2026. Start ≤ current year, End > current year	Drives alumni transition trigger. Diploma = 3yr, UG = 3–4yr, PG = 1–2yr
Registration / Roll Number	Text	Required	Alphanumeric, 4–20 chars. Must be unique per college	Used to prevent duplicate accounts at same college. Not public.
Hubbler Username (Handle)	Text	Required	Lowercase letters, numbers, dots only. 3–30 chars. Globally unique.	System suggests @first name.collegecode. User can edit. Shown as @handle everywhere.
Email Address	Email	Required	Valid email format. Unique across all users.	Magic link sent here. Student college email preferred but not enforced.
Password	Password	Required	Min 8 chars. Must contain 1 uppercase, 1 number, 1 special char.	Bcrypt hashed. Never stored plain. Used as signup-time credential.
Profile Photo	Image Upload	Optional	JPG/PNG, max 5MB. Resized to 400x400 on server.	Can be added later from profile settings.

1.2 Auto-Generated Fields (not shown to user)

FIELD	TYPE	REQ	VALIDATION	NOTES
System ID	String	Auto	Format: HX-{YYYY}-{5-digit zero-padded seq}. e.g. HX-2025-00001	PERMANENT. Never changes. Printed on QR, certs.
Account Status	Enum	Auto	ACTIVE_STUDENT ALUMNI HIGHER_STUDIES_PENDING	Set to ACTIVE_STUDENT on signup
Annual Credits	Integer	Auto	Default 0. Resets to 0 on June 1 each year via cron job.	Drives leaderboard rank badge
Lifetime Credits	Integer	Auto	Default 0. Never resets. Increments only.	Shown permanently on profile even as alumni
Created At / Updated At	Timestamp	Auto	UTC timestamps set by DB on insert/update	Standard audit fields

1.3 Student Signup Flow

Step 1	Student fills signup form on /signup/student
Step 2	Client-side validation fires on all fields before submit
Step 3	Server validates: email uniqueness, handle uniqueness, roll number uniqueness within college, college exists in DB
Step 4	If valid: create user record (status = PENDING_EMAIL_VERIFY), generate HX System ID, send magic link email
Step 5	Email arrives: 'Confirm your HubblerX account' — contains secure token link (expires 24 hours)
Step 6	Student clicks link → /auth/verify?token=XXXX → account status = ACTIVE_STUDENT, session created, redirect to /onboarding
Step 7	Onboarding screen shows: Hubbler ID card, prompt to add profile photo, quick tour of leaderboard

1.4 Student Login Flow

Students have two login options:

- **Option A — Password login:** Email + password on /login. On success: JWT session + redirect to /dashboard/student
- **Option B — Magic link login:** Student enters email only on /login/magic. Server generates token, sends "Sign in to HubblerX" email. Student clicks link, session created, redirect to /dashboard/student. Token expires in 15 minutes. Single-use.

NOTE
Magic link is the preferred login method. Promote it as the default. Password login is the fallback. Never store magic link tokens in plain text — use a secure hash in the DB and compare on verify.

1.5 Handle Suggestion Logic

When the user types their name and selects their college, the system should auto-suggest a handle in the format `firstname.collegeshortcode`, e.g. `arjun.ssn`. If taken, append a number: `arjun.ssn2`. Shortcodes for colleges are set when the college registers. The user can override the suggestion freely.

Example college shortcodes: `ssn` · `ceg` · `vit` · `srm` · `anna` · `psg` · `nec` · `sece` · `kce`

College Admin Signup & Login

College admins are the supply side of HubblerX — they create events, manage check-in, and select winners. The college signup is a separate, more formal flow that requires manual verification by the Support team before the account goes live. A college with multiple campuses (e.g. SRM Kattankulathur vs SRM Vadapalani) registers each campus as a separate College Admin account.

2.1 College Admin Signup Form — All Fields

FIELD	TYPE	REQ	VALIDATION	NOTES
College Official Name	Text	Required	2–120 chars. Must match institution's official registered name.	Shown on all events, community page, and certificates.
Campus / Branch Name	Text	Conditional	Required only if college has multiple campuses. Free text.	e.g. 'Kattankulathur Campus', 'Vadapalani Campus'. Appended to college name in display.
College Location (City)	Select	Required	Dropdown: Chennai Coimbatore Madurai Trichy Other	Drives city-specific leaderboard and discovery.
Specific Area / District	Text	Required	Free text area/locality within city. e.g. 'Kattankulathur', 'Vadapalani', 'Guindy'	Helps students identify the right campus when multiple exist.
Full Street Address	Textarea	Required	Standard address string. Min 10 chars.	Shown on event listings so students know where to go.
Google Maps Link	URL	Optional	Must be a maps.google.com or goo.gl/maps URL.	Embedded on event pages for navigation.
Admin Full Name	Text	Required	Name of the person creating this admin account.	Could be DSW, student affairs officer, or committee lead.
Admin Role / Designation	Select	Required	DSW Student Affairs Officer Faculty Coordinator Student Committee Lead Other	For our records only. Not shown publicly.
Official College Email	Email	Required	Must match college domain (e.g. @ssn.edu.in, @srmist.edu.in). Validated against college name.	Primary login email. Magic link sent here. Cannot be a personal Gmail/Yahoo.
Mobile Number	Phone	Required	10 digits, India. OTP verified.	For urgent support communication only.
Password	Password	Required	Min 8 chars. 1 uppercase, 1 number, 1 special char.	Bcrypt hashed.
College Logo	Image Upload	Optional	PNG/JPG, max 5MB, min 200x200px. Stored as square crop.	Appears on community page, event pages, and certificates.

FIELD	TYPE	REQ	VALIDATION	NOTES
College Shortcode	Text	Required	2–8 lowercase letters. e.g. 'ssn', 'srmktr', 'ceg'. Globally unique.	Used in student handle suggestions: @arjun.ssn. Set once, cannot change.
Departments Offered	Multi-Select	Required	Select from standard dept list + custom add. Min 1.	Populates the dept dropdown in student signup for this college.
Education Tiers Offered	Multi-Select	Required	Select all that apply: Diploma UG PG	Determines which student tier options appear for this college.
NAAC/AICTE Reference	Text	Optional	Accreditation ID if available.	Used by Support team during manual verification to confirm college is real.

2.2 College Admin Signup Flow

Step 1	Admin fills /signup/college form and submits.
Step 2	Server validates: official email domain, shortcode uniqueness, all required fields.
Step 3	Account created with status = PENDING_VERIFICATION. Admin sees: 'Your application is under review. You will receive an email within 24–48 hours.'
Step 4	Support team receives notification in the Support dashboard: new college pending verification. They review NAAC reference, email domain, and college name. One-click Approve or Reject with reason.
Step 5	On Approve: status = ACTIVE. Verification confirmation email sent to admin with magic sign-in link. College now appears in student signup dropdown.
Step 6	On Reject: rejection email sent with reason. Admin can resubmit with corrections.
Step 7	Admin clicks magic link, session created, redirected to /dashboard/college — the college admin panel.

2.3 Multi-Campus Handling

For universities with multiple campuses (SRM, VIT, Anna University affiliated colleges, etc.):

- Each campus registers as a separate College Admin account with its own email, shortcode, and admin profile.
- The college official name is shared (e.g. 'SRM Institute of Science and Technology') but the campus name differentiates them (e.g. 'Kattankulathur Campus').
- In student-facing dropdowns, display as: 'SRM Institute of Science and Technology — Kattankulathur Campus'
- Students pick their specific campus at signup. Their Hubbler ID, leaderboard position, and community page are scoped to that specific campus, not the parent university.
- Support can link campus accounts to a parent institution in the backend for analytics, but this is an internal grouping only.

CRITICAL

Never let a college admin see data from another campus — even if it's the same university. Each campus account is completely isolated. Data isolation is enforced at the query level, not just the UI level.

Support Role — Login & Provisioning

Support accounts are for the HubblerX team only. There is no public signup page for Support. Support accounts are created by an existing Support account (or by a seed script on first deploy). Support users log in with their own email and password — no magic link is needed since these are internal accounts.

3.1 Support Login

- URL: /admin/login — this route must NOT be linked from anywhere on the public site. It is accessed by direct URL only.
- Login form: Email address + Password only. No magic link flow for Support.
- On success: JWT with role = SUPPORT, redirect to /admin/dashboard.
- Enforce 2-Factor Authentication (TOTP via Google Authenticator) for all Support logins. This is mandatory, not optional.
- Session expiry: 8 hours. Support users must re-authenticate after 8 hours regardless of activity.
- Failed login attempts: lock account after 5 consecutive failures. Unlock only via another Support account.

3.2 Support Account Provisioning

Support accounts are created by an existing Support admin through the /admin/team panel. Fields required:

FIELD	TYPE	REQ	VALIDATION	NOTES
Full Name	Text	Required	2–60 chars	Shown in audit logs when this user takes actions
Email Address	Email	Required	Unique across all users. Preferably a @hubblerx.in email.	Used as login credential
Password	Password	Required	Min 12 chars for Support accounts. Strong policy enforced.	Must be changed on first login.
Access Level	Select	Required	SUPPORT_ADMIN SUPPORT_MODERATOR	Admin: full platform control. Moderator: content moderation only.
2FA Setup	TOTP	Required	Must enroll TOTP app before account is activated.	QR code shown on first login. Cannot skip.

3.3 Support Admin vs Support Moderator

CAPABILITY	SUPPORT ADMIN	SUPPORT MODERATOR
Approve/reject college signup	✓	✗
Create Support team accounts	✓	✗
Override credits with audit note	✓	✗
Manage annual leaderboard reset	✓	✗
Moderate flagged community posts	✓	✓
Resolve marketplace disputes	✓	✓
View full audit log	✓	■ Read
View platform analytics	✓	✓

Select Featured Project of Week	✓	✓
Ban/suspend user accounts	✓	X

Authentication Architecture

4.1 Token Strategy

- **JWT (JSON Web Token)** for all session management. Include in Authorization header as Bearer token. Never store in localStorage — use httpOnly cookies.
- **JWT payload:** { userId, role, collegeld (if admin), sessionId, iat, exp }
- **Token expiry:** Student = 7 days. College Admin = 24 hours. Support = 8 hours.
- **Refresh tokens:** Stored in httpOnly cookie. Student refresh = 30 days. Admin = 7 days. Support = no refresh — must re-login.
- **Magic link tokens:** Stored as bcrypt hash in DB. Student login links expire in 15 minutes. Signup verify links expire in 24 hours. Single-use — mark as used immediately on verify.

4.2 Route Protection Rules

/dashboard/student/*	JWT required, role = STUDENT, status = ACTIVE_STUDENT
/dashboard/college/*	JWT required, role = COLLEGE_ADMIN, status = ACTIVE
/admin/*	JWT required, role = SUPPORT_ADMIN or SUPPORT_MODERATOR. 2FA verified flag must be true in token.
/events/[id]	Public route — no auth needed to view event page or buy ticket. Auth needed for logged-in experience.
/profile/[handle]	Public route — profiles are public. Internship apply button requires auth.
/leaderboard	Public route — visible without login. No personal data shown.
/signup/*	Redirect to dashboard if already authenticated (avoid double signup).
/admin/login	Not linked from public nav. Rate limit: max 5 attempts per 15 min per IP.

WARNING
ALUMNI accounts (status = ALUMNI) can only access: their own public profile, marketplace sell-only (30-day window), and read-only leaderboard. Block all other authenticated routes — return 403 with message 'This feature is available for active students only.'

Role-Based Access Control — Full Feature Matrix

This is the definitive permission matrix. Every feature on the platform maps to this table. When building a component or API endpoint, find the feature here and implement exactly the permission shown. If a feature is not in this table, treat it as NONE for all roles until it is added.

5.1 Authentication & Profile

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
Create a student account (signup)	✓ Full	✗ None	✓ Full
Create a college admin account (signup)	✗ None	✓ Full	✓ Full
Log in with email + password	✓ Full	✓ Full	✓ Full
Log in with magic link (passwordless)	✓ Full	✓ Full	✗ None
View own Hubbler profile	✓ Full	✗ None	✓ Full
Edit own profile (name, photo, bio)	✓ Full	✗ None	✗ None
View another student's public profile	✓ Full	✓ Full	✓ Full
Change password	✓ Full	✓ Full	✓ Full
Deactivate own account	✓ Full	✗ None	✓ Full
Ban / suspend any account	✗ None	✗ None	✓ Full
View alumni profiles (read-only)	✓ Full	✓ Full	✓ Full

5.2 Event Ticketing

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
Browse event discovery feed	✓ Full	✓ Full	✓ Full
View event detail page	✓ Full	✓ Full	✓ Full
Purchase a ticket (UPI)	✓ Full	✗ None	✗ None
View own purchased tickets	✓ Full	✗ None	✗ None
Download QR ticket PDF	✓ Full	✗ None	✗ None
Create a new event	✗ None	■ Own college	✓ Full
Edit own college's event	✗ None	■ Own college	✓ Full
Delete / cancel an event	✗ None	■ Own college	✓ Full
Use QR check-in scanner	✗ None	■ Own college	✓ Full
View live ticket sales dashboard	✗ None	■ Own college	✓ Full
Select winners and confirm results	✗ None	■ Own college	✓ Full
Override winner selection (dispute)	✗ None	✗ None	✓ Full
Download attendee list (.xlsx)	✗ None	■ Own college	✓ Full
Generate event poster (auto)	✗ None	■ Own college	✓ Full
View payout statement	✗ None	■ Own college	✓ Full

5.3 Credits & Leaderboard

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
View city-wide leaderboard	✓ Full	✓ Full	✓ Full
View college-wise leaderboard	✓ Full	✓ Full	✓ Full
View dept-wise leaderboard	✓ Full	✓ Full	✓ Full
View tech / non-tech split boards	✓ Full	✓ Full	✓ Full
View own annual + lifetime credits	✓ Full	✗ None	✓ Full
Credits auto-assigned after win	✓ Full	✗ None	✗ None
Raise a credit dispute (48hr window)	✓ Full	✗ None	✗ None
View credit dispute queue	✗ None	✗ None	✓ Full
Override / correct credit assignment	✗ None	✗ None	✓ Full
View immutable credit audit log	✗ None	✗ None	✓ Full
Trigger annual leaderboard reset	✗ None	✗ None	✓ Full

5.4 Community Pages

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
View any college's community page	✓ Full	✓ Full	✓ Full
Post to own college community page	✓ Full	■ Own college	✗ None
Post official announcement	✗ None	■ Own college	✗ None
Pin post to top (max 3)	✗ None	■ Own college	✓ Full
Delete own post	✓ Full	■ Own college	✗ None
Delete any post (moderation)	✗ None	✗ None	✓ Full
Flag / report a post	✓ Full	✓ Full	✗ None
View flagged posts queue	✗ None	✗ None	✓ Full
Auto-post: event published	✗ None	✗ None	✗ None
Auto-post: winner confirmed	✗ None	✗ None	✗ None
Reshare a community post	✓ Full	✓ Full	✗ None
Select Featured Project of Week	✗ None	✗ None	✓ Full

5.5 Student Marketplace

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
Browse marketplace listings	✓ Full	✗ None	✓ Full
Create a listing (sell item)	✓ Full	✗ None	✓ Full
Edit own listing	■ Own only	✗ None	✓ Full
Delete own listing	■ Own only	✗ None	✓ Full
Message a seller (in-platform chat)	✓ Full	✗ None	✗ None

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
View seller Hubbler profile	✓ Full	✗ None	✓ Full
Leave seller rating after transaction	✓ Full	✗ None	✗ None
Report a listing	✓ Full	✗ None	✗ None
View reported listings queue	✗ None	✗ None	✓ Full
Resolve marketplace dispute	✗ None	✗ None	✓ Full
Alumni: sell items (30-day window)	■ Partial	✗ None	✗ None
Alumni: buy items	✗ None	✗ None	✗ None

5.6 Internship Board

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
Browse internship listings	✓ Full	✗ None	✓ Full
Apply to an internship (one tap)	✓ Full	✗ None	✗ None
View own applications	✓ Full	✗ None	✓ Full
Post an internship listing (startup)	✗ None	✗ None	✓ Full
Approve startup verification	✗ None	✗ None	✓ Full
Browse student profiles (talent search)	✗ None	✗ None	✓ Full
Message an applicant	✗ None	✗ None	✗ None
View application inbox (startup)	✗ None	✗ None	✓ Full

NOTE

Startups are a 4th actor in the internship module — they are not one of the 3 core roles. A startup admin account (STARTUP role) needs to be provisioned by Support. Startup accounts have access only to: post listings, view applicants, message applicants. No other platform access.

5.7 Project Showcase

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
Browse project showcase (all)	✓ Full	✓ Full	✓ Full
Upload a project	✓ Full	✗ None	✓ Full
Edit own project	■ Own only	✗ None	✓ Full
Delete own project	■ Own only	✗ None	✓ Full
Tag teammates on project	✓ Full	✗ None	✗ None
Accept/reject teammate tag	✓ Full	✗ None	✗ None
Link project to event win (verified)	✓ Full	✗ None	✗ None
Star / save a project	✓ Full	✓ Full	✓ Full
Comment on a project	✓ Full	✓ Full	✗ None
Report a project	✓ Full	✓ Full	✗ None
Feature project of the week	✗ None	✗ None	✓ Full

5.8 LinkedIn Certificates

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
View own certificates	✓ Full	✗ None	✓ Full
Download certificate as PDF	✓ Full	✗ None	✓ Full
Share certificate to LinkedIn (deep link)	✓ Full	✗ None	✗ None
View verification URL (public)	✓ Full	✓ Full	✓ Full
Auto-generate cert after event result	✗ None	✗ None	✗ None
Revoke a certificate (dispute resolved)	✗ None	✗ None	✓ Full

5.9 College Admin Dashboard

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
View event centre (own events)	✗ None	■ Own college	✓ Full
View student directory (own college)	✗ None	■ Own college	✓ Full
Download student directory (.xlsx)	✗ None	■ Own college	✓ Full
View leaderboard (own college)	✗ None	■ Own college	✓ Full
Download leaderboard (.xlsx)	✗ None	■ Own college	✓ Full
Download semester summary PDF	✗ None	■ Own college	✓ Full
Download attendee list per event	✗ None	■ Own college	✓ Full
Download ticket sales report	✗ None	■ Own college	✓ Full
Download winners + credits report	✗ None	■ Own college	✓ Full
Download dept performance report	✗ None	■ Own college	✓ Full
Download alumni report	✗ None	■ Own college	✓ Full
Schedule automated report delivery	✗ None	■ Own college	✓ Full
View payout history	✗ None	■ Own college	✓ Full
View another college's data	✗ None	✗ None	✓ Full

5.10 Support / Platform Admin Panel

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
View pending college verifications	✗ None	✗ None	✓ Full
Approve college signup	✗ None	✗ None	✓ Full
Reject college signup (with reason)	✗ None	✗ None	✓ Full
View all colleges + their stats	✗ None	✗ None	✓ Full
View all students (platform-wide)	✗ None	✗ None	✓ Full
View platform-wide analytics	✗ None	✗ None	✓ Full
Manage support team accounts	✗ None	✗ None	✓ Full
Trigger annual credit reset (June 1)	✗ None	✗ None	✓ Full

FEATURE / SCREEN	STUDENT	CLG ADMIN	SUPPORT
View + action alumni transition queue	✗ None	✗ None	✓ Full
View full platform audit log	✗ None	✗ None	✓ Full

Screen-by-Screen Feature Ownership

Every page of the HubblerX web app and its primary owner role. Build screens in priority order — P0 first.

URL	SCREEN	ACCESS	PRI	NOTES
/signup/student	Student Signup	Public	P0	Multi-step form. Steps: Personal Info → College Info → Account Setup → Email Verify
/signup/college	College Admin Signup	Public	P0	Single long-form page with grouped sections. After submit: pending state screen.
/login	Login (Student)	Public	P0	Email + password OR magic link option. Redirect by role on success.
/admin/login	Support Login	Internal	P0	Email + password + TOTP. Dark page, no nav. Not in sitemap.
/auth/verify	Email Verify Handler	System	P0	No UI — handles token, sets session, redirects. Shows error if token invalid/expired.
/onboarding	Student Onboarding	Student	P0	Hubbler ID reveal screen. Profile photo prompt. 3-step tour. Skip option.
/dashboard/student	Student Home	Student	P0	Feed: upcoming events, recent wins at college, leaderboard position card, marketplace highlights.
/dashboard/student/profile	Hubbler Profile	Student	P0	Public-facing profile: handle, rank, credits, event history, projects, certificates.
/dashboard/student/leaderboard	Leaderboard	Student	P0	Tabs: City College Dept. Toggle: Tech / Non-tech. Tier selector: Diploma / UG / PG.
/events	Event Discovery	Public	P0	Grid of upcoming events. Filter by college, type, date. Search.
/events/[id]	Event Page	Public	P0	Event details, ticket tiers, buy button. Share link. Countdown.
/events/[id]/buy	Ticket Purchase	Student	P0	Tier select → Razorpay UPI → success screen with QR ticket.
/dashboard/college	College Admin Home	College Admin	P0	Stats overview: events live, tickets sold today, recent check-ins, pending results.
/dashboard/college/events/new	Create Event	College Admin	P0	Full event creation form. Auto-poster preview. Publish button.
/dashboard/college/events/[id]/scanner	QR Scanner	College Admin	P0	Full-screen PWA scanner. Offline-capable. Green/red result. Attendance counter.
/dashboard/college/events/[id]/results	Winner Selection	College Admin	P0	Per-category dropdowns. Confirm results button. Auto-post preview before confirm.
/dashboard/college/reports	Download Reports	College Admin	P1	Report type selector + date range + dept filter + format (.xlsx/.csv/.pdf). Download button.
/dashboard/college/students	Student Directory	College Admin	P1	Searchable table of all registered students at this college.
/community/[collegeCode]	College Community Page	Public	P1	Chronological feed. Pinned posts at top. Create post button for verified students.
/marketplace	Marketplace Browse	Student	P1	Grid of listings. Filter: category, college-first, price. My listings tab.

URL	SCREEN	ACCESS	PRI	NOTES
/marketplace/new	Create Listing	Student	P1	Category select → fields → photos → price → condition → preview → publish.
/projects	Project Showcase	Public	P1	Browse all projects. Filter by dept, category, college, has-win-badge.
/projects/new	Upload Project	Student	P1	Title, description, category, tags, file upload, video link, team tags, event link.
/internships	Internship Board	Public	P2	Browse verified startup listings. Filter by role, dept, stipend, mode.
/certificates/[id]	Certificate View + Share	Student	P1	Certificate render + Download PDF + Share to LinkedIn button.
/verify/[eventId]/[systemId]	Certificate Verification	Public	P1	Public page. No login. Shows: name, event, position, date, green Verified badge.
/admin/dashboard	Support Dashboard	Support	P0	Platform stats. Pending verifications count. Flagged content count. Quick links.
/admin/colleges	Manage Colleges	Support	P0	All colleges list. Pending verification queue. Approve/reject actions.
/admin/credits	Credit Audit Log	Support	P1	Immutable log of all credit actions. Filter by student, event, date. Override form.
/admin/disputes	Disputes Queue	Support	P1	Open disputes: credit disputes + marketplace disputes. Action panel.
/admin/moderation	Content Moderation	Support	P1	Flagged posts queue. Approve / delete actions.
/admin/team	Support Team Management	Support Admin	P1	List support accounts. Create new support user form.
/admin/leaderboard-reset	Annual Reset Control	Support Admin	P1	Preview of what changes. Confirm reset button. Only available June 1–7.

Database Schema Notes

These are not full migrations — they are the key fields and constraints the dev team must plan for. Use these as the source of truth for table design.

TABLE: users

- id UUID PK
- system_id VARCHAR(20) UNIQUE NOT NULL — HX-YYYY-NNNNN
- handle VARCHAR(30) UNIQUE NOT NULL — @handle
- full_name VARCHAR(60) NOT NULL
- email VARCHAR(255) UNIQUE NOT NULL
- password_hash VARCHAR(255) NOT NULL
- mobile VARCHAR(15) NOT NULL
- role ENUM('STUDENT','ALUMNI','STARTUP') DEFAULT 'STUDENT'
- status ENUM('PENDING_EMAIL_VERIFY','ACTIVE_STUDENT','ALUMNI','SUSPENDED')
- college_id UUID FK → colleges.id
- department VARCHAR(60)
- education_tier ENUM('DIPLOMA','UG','PG')
- start_year SMALLINT
- end_year SMALLINT
- roll_number VARCHAR(20) — unique per college (enforce via UNIQUE(college_id, roll_number))
- annual_credits INTEGER DEFAULT 0
- lifetime_credits INTEGER DEFAULT 0
- created_at TIMESTAMP DEFAULT NOW()
- updated_at TIMESTAMP

TABLE: colleges

- id UUID PK
- official_name VARCHAR(120) NOT NULL
- campus_name VARCHAR(80) — nullable, for multi-campus
- shortcode VARCHAR(8) UNIQUE NOT NULL
- city VARCHAR(60) NOT NULL
- area VARCHAR(80) NOT NULL
- full_address TEXT
- maps_link VARCHAR(255)
- admin_email VARCHAR(255) UNIQUE NOT NULL
- admin_name VARCHAR(60) NOT NULL
- admin_designation VARCHAR(60)
- mobile VARCHAR(15)
- logo_url VARCHAR(255)
- departments TEXT[] — array of department names
- education_tiers TEXT[] — ['UG','PG'] etc
- naac_ref VARCHAR(60)
- status ENUM('PENDING_VERIFICATION','ACTIVE','SUSPENDED')
- verified_by UUID FK → support_users.id — nullable
- verified_at TIMESTAMP
- created_at TIMESTAMP DEFAULT NOW()

TABLE: support_users

- id UUID PK
- full_name VARCHAR(60) NOT NULL
- email VARCHAR(255) UNIQUE NOT NULL
- password_hash VARCHAR(255) NOT NULL
- totp_secret VARCHAR(64) — encrypted at rest
- totp_enrolled BOOLEAN DEFAULT FALSE
- access_level ENUM('SUPPORT_ADMIN','SUPPORT_MODERATOR')
- created_by UUID FK → support_users.id — who provisioned this account
- last_login TIMESTAMP
- is_active BOOLEAN DEFAULT TRUE
- created_at TIMESTAMP DEFAULT NOW()

TABLE: magic_link_tokens

- id UUID PK
- user_id UUID FK → users.id OR colleges.id — polymorphic
- user_type ENUM('STUDENT','COLLEGE_ADMIN')
- token_hash VARCHAR(255) NOT NULL — bcrypt hash of the actual token
- purpose ENUM('SIGNUP_VERIFY','LOGIN') NOT NULL
- expires_at TIMESTAMP NOT NULL
- used_at TIMESTAMP — null until used. Set immediately on use.
- created_at TIMESTAMP DEFAULT NOW()

TABLE: events

- id UUID PK
- college_id UUID FK → colleges.id
- title VARCHAR(120) NOT NULL
- event_type ENUM('HACKATHON','CODING','CULTURAL','SPORTS','WORKSHOP','SEMINAR','OTHER')
- is_tech BOOLEAN — derived from event_type, stored for query performance
- date_start TIMESTAMP NOT NULL
- date_end TIMESTAMP
- venue VARCHAR(200)
- max_participants INTEGER
- status ENUM('DRAFT','LIVE','CHECKIN_OPEN','COMPLETED','CANCELLED')
- poster_url VARCHAR(255)
- credit_multiplier NUMERIC(3,1) DEFAULT 1.0 — 1.5 for tech events
- created_at TIMESTAMP DEFAULT NOW()

TABLE: credit_audit_log

- id UUID PK
- user_id UUID FK → users.id
- event_id UUID FK → events.id — nullable for manual overrides
- credits_awarded INTEGER NOT NULL
- credit_type ENUM('PARTICIPATION','3RD','2ND','1ST','MANUAL_OVERRIDE','RESET')
- awarded_by_role ENUM('SYSTEM','SUPPORT') NOT NULL
- awarded_by_id UUID — support_user id if manual, null if system
- override_reason TEXT — required for MANUAL_OVERRIDE
- created_at TIMESTAMP DEFAULT NOW() — IMMUTABLE: no update/delete allowed on this table

Email Flow Specifications

Student Signup — Verify Email

Token type: SIGNUP_VERIFY	Expiry: 15 min from send	Subject: "Verify your HubblerX account, [Name]"
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- CTA button: 'Confirm Email & Create My Hubbler ID'
- Link: /auth/verify?token=XXXX&type=signup
- Plain text fallback link included
- If not you? link at bottom
- Expires notice: 'This link expires in 24 hours'

Student Magic Link Login

Token type: LOGIN	Expiry: 15 min	Subject: "Your HubblerX sign-in link"
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- CTA: 'Sign In to HubblerX'
- Link: /auth/verify?token=XXXX&type=login
- Security note: 'If you did not request this, you can ignore this email.'
- Link expires in 15 minutes — show this clearly
- Single-use: clicking twice shows 'link already used' error

College Admin — Signup Under Review

Token type: —	Expiry: Instant	Subject: "We received your HubblerX college application"
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- No link in this email — just confirmation
- State: 'We are reviewing your application. You will hear from us within 24–48 hours.'
- Show the college name they registered with

College Admin — Approved

Token type: LOGIN	Expiry: On approve	Subject: "Your college is now live on HubblerX — [College Name]"
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- CTA: 'Access Your College Dashboard'
- Magic sign-in link (expires 48 hours for this one)
- Include college shortcode and a note to share with their committee

College Admin — Rejected

Token type: —	Expiry: On reject	Subject: "Your HubblerX college application — action required"
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- Show the rejection reason (entered by Support)
- CTA: 'Resubmit Application' → /signup/college with email pre-filled
- Support contact email for questions

Win Notification to Student

Token type: —	Expiry: On result confirm	Subject: "You placed [Position] at [Event Name]! ■"
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- Show credits earned: +100 credits
- CTA: 'View Your Certificate' → /certificates/[id]
- Secondary CTA: 'Share to LinkedIn'
- Show new leaderboard rank if it changed

Alumni Transition Warning

Token type: —	Expiry: 60 days before end_year	Subject: "Your HubblerX account is changing — action needed"
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- Explain: graduation year is approaching
- Two CTA buttons: 'Continue to Higher Studies' | 'I'm Graduating'
- 60-day deadline shown clearly
- What happens if no action: auto-convert to Alumni after end_year

Credit Dispute Filed

Token type: —	Expiry: On dispute submit	Subject: "Your credit dispute has been received"
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- Dispute reference number
- 48-hour resolution window mentioned
- What to expect: Support team reviews and either corrects or upholds the result

Validation Rules & Error States

9.1 Client-Side Validation (fire on blur + on submit)

FIELD	RULE
handle	Lowercase only, letters/numbers/dots, 3–30 chars, no consecutive dots, no leading/trailing dot. Regex: <code>^[a-z0-9][a-z0-9.]{1,28}[a-z0-9]\$</code> — Check uniqueness via debounced API call (400ms delay). Show green tick when available.
email	Valid format (RFC 5322 basic). Uniqueness checked on blur via API. College admin email must match a known college domain suffix.
mobile	Exactly 10 digits after stripping +91 prefix. No spaces/hyphens. OTP sent to verify at signup.
password (student)	Min 8 chars. Strength meter: Weak (red) < 8 / Medium (amber) has upper+number / Strong (green) has upper+number+special.
password (support)	Min 12 chars. Must have uppercase, number, special char. No dictionary words. Reject common passwords list.
year range	Start year: current year minus 4 ≤ start ≤ current year. End year: start + tier_duration (Diploma=3, UG=4, PG=2). Show derived graduation year prominently.
roll number	Alphanumeric, 4–20 chars. Unique per college — check via API. Case-insensitive normalise to uppercase before storing.
college shortcode	2–8 lowercase letters only (a-z). Globally unique. Suggest auto from college name initials. Cannot contain numbers.
campus/branch name	Only shown when college name matches a known multi-campus institution. 2–80 chars. Free text.

9.2 Critical Error States to Handle

Duplicate email at signup	Show inline error: 'An account with this email already exists. Sign in instead?' with link.
Duplicate handle	Real-time check: 'This username is taken. Try @arjun.ssn2?' with auto-suggestion.
College not found in DB	Student selects college not in system: 'Your college isn't on HubblerX yet. Let them know!' — share link to college signup page.
Magic link expired	Clear error: 'This sign-in link has expired. Request a new one.' — one-click resend.
Magic link already used	'This link has already been used. If you need to sign in, request a new link.' Do not indicate whether account exists.
College signup pending verification	Any login attempt by pending college admin: 'Your application is under review. You will be notified by email.'
Support TOTP fail	After 3 wrong TOTP codes: show lockout warning. After 5: lock account, alert other Support admins via email.
Alumni tries to access student feature	403 page with message: 'This feature is only available for active students.' No further detail.

Roll number already registered at college	Inline error on field: 'A student with this roll number is already registered at [College]. If this is you, sign in instead.'
College email domain mismatch	For college admin: 'Please use your official college email address (e.g. name@srmist.edu.in)'

HubblerX Developer Specification — End of Document

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